AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 29 · STUDENT LESSON

Find a Real Problem

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Week 23: evaluate success using recorded cases.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Describe a specific user difficulty using supplied evidence without inventing its cause. Try the recall before opening notes.

Your learning target

I can describe a specific **user** difficulty using supplied **evidence** without inventing its cause.

I can define a useful **outcome** and a way to observe it before choosing a solution.

I can choose a small project with clear boundaries, permitted information and a simple alternative.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Humans Set Boundaries. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Prompts Are Design Instructions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Fair for Whom?. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A strong project begins with a specific **user** and observable **need**.



Your target

LEARNING TARGET Students spend three minutes finding returned books. Compare shelf labels with an AI locator. Define one fair success test before choosing a solution.

Learner name:

Readiness for the connected explanation

A teacher permits a paper prototype using fictional activity requests. Does this permit collecting real pupils' names and publishing their choices? Explain.

This week's end goal

By the end, I can turn a recorded problem into a bounded, measurable project goal.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Find a Real Problem case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY 'Students **need** AI' is not a **problem statement**; 'students spend five minutes finding misplaced supplies after art' is observable and testable.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

For the final project sequence, we will develop a small activity recommender on paper. Its **user**s are fictional pupils choosing a classroom activity that fits their available time and room conditions.

We will decide what the recommender must do, build a traceable **process**, test it and explain its limits. A recommendation is a suggestion; the pupil can decline it.

Start before the solution. Our supplied **observation** log describes ten fictional requests made using an activity catalogue. Six first choices fit the request's time and room conditions. Four did not and **need**ed another choice.

These invented classroom records give us a problem to investigate. They are not **observation**s of your classmates and do not show how often this happens in other classes.

A useful **problem statement** is: "In the supplied ten-request exercise, four first choices did not fit the stated time or room conditions, so those **user**s had to choose again." "We **need** AI" names a possible technology but says neither who **need**s help nor what should improve.

The log also does not establish whether unclear wording, missing information or another cause produced the mismatches.

STUDENT LESSON · 3 OF 18

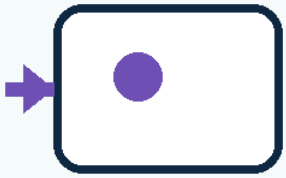
See how it works

> Follow the mechanism for Find a Real Problem. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.



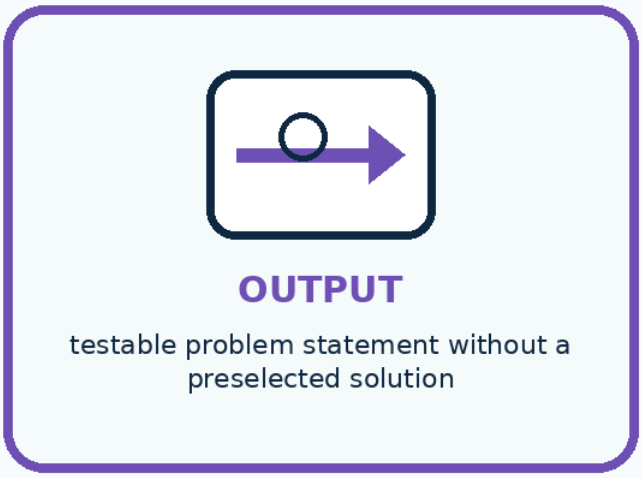
INPUT

observations and interview evidence



PROCESS

identify user, situation, need, and measurable effect



INPUT observations and **interview evidence**

PROCESS identify **user**, situation, **need**, and measurable effect

OUTPUT testable **problem statement** without a preselected solution

Key idea

CORE IDEA A strong project begins with a specific **user** and observable **need**.

How the mechanism works

UNDER THE HOOD Problem framing separates **evidence** about a **user**'s **need** from assumptions about a solution. A strong statement identifies who, when, what observable difficulty occurs, and how improvement could be measured.

Draw a fresh example with all three stages:

A closer explanation

From an observed difficulty to a testable goal

EVIDENCE OR DECISION	WHAT IT SUPPORTS	WHAT REMAINS TO CHECK
4 of 10 choices mismatch	Some re-selection occurred	Cause is not established
Need a compatible choice	Check time and conditions	User can still decline
Paper suggestion process	Small permitted prototype	Benefit needs a comparison
Measure fit and coverage	Observe useful outcomes	Do not hide no-result cases

The goal describes a useful result. Building a tool is an activity, not evidence that users benefited.

Our target **outcome** is a compatible suggestion when the catalogue contains one, and a clear no-match message when none fits.

Compatibility means respecting the stated conditions; it does not mean that every pupil will enjoy the activity. We can check time and room conditions from a request and catalogue. Enjoyment would **need** different **evidence**.

Count compatible suggestions, incompatible suggestions and no-match messages separately. Otherwise a system could avoid every wrong suggestion by refusing to suggest anything.

If a valid option existed, that refusal would still fail to provide the intended help. Record the total number of requests so a count has a clear denominator.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

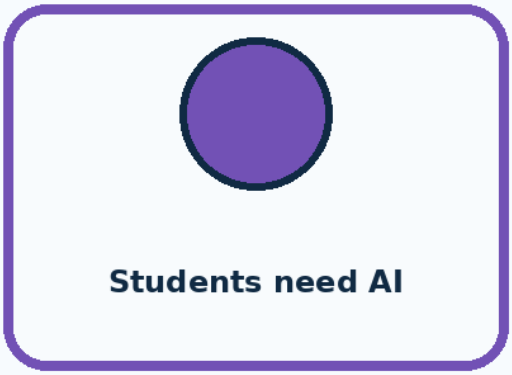
Compare the cases

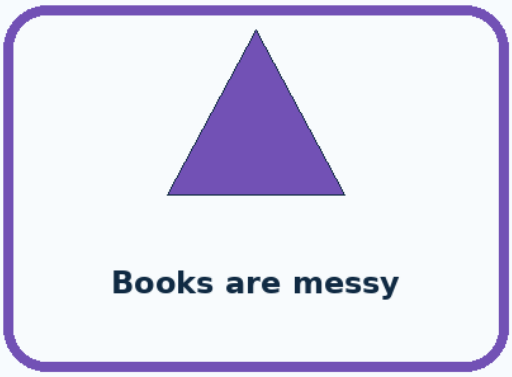
> Compare the supplied cases for Find a Real Problem. Identify the detail that changes the conclusion.

PICTURE GALLERY

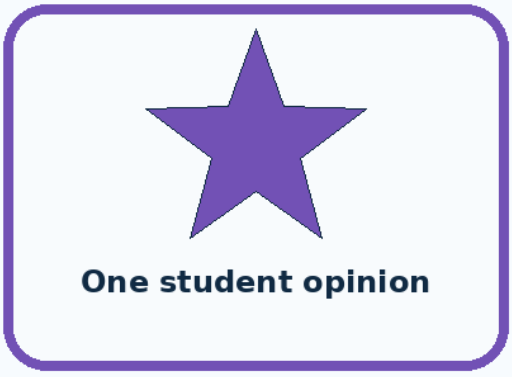
Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









The case to investigate

CASE 1 — Students **need** AI Clue: technology-first claim. Meaning: weak framing.

The case to investigate

CASE 2 — Books are messy Clue: vague judgment. Meaning: **need**s **observation**.

The case to investigate

CASE 3 — Librarian re-sorts 20 books daily Clue: specific burden. Meaning: testable **need**.

The case to investigate

CASE 4 — One student opinion Clue: useful but limited. Meaning: seek more **evidence**.

Keep this boundary clear

BOUNDARY Do not collect personal stories unrelated to the design question.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: The approved list contains no tray with every requested tool.

Predicted result

Prediction: The finder should report no suitable tray rather than guess.

Why the result follows

Reason: Success includes honestly identifying unsatisfied requirements.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Find a Real Problem to explain a concrete decision, rather than only memorising definitions.

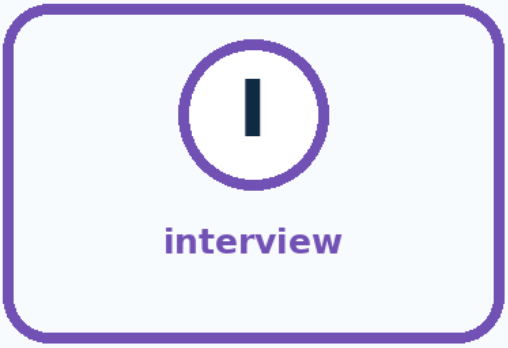
VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
user	person affected by the problem or solution	
need	observable difficulty or goal	

Word	Meaning	My example
interview	structured questions used to learn from a person	
success measure	evidence showing whether the need improved	

Misconception detective

> **Find the unsupported claim**
>
> CLAIM TO REPAIR A popular technology is proof of a real problem.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
User need	Something a person needs to accomplish in a particular situation
Problem statement	A specific description of who faces a difficulty and what that difficulty is
Outcome	The result experienced by the user
Constraint	A condition that a proposed solution must respect

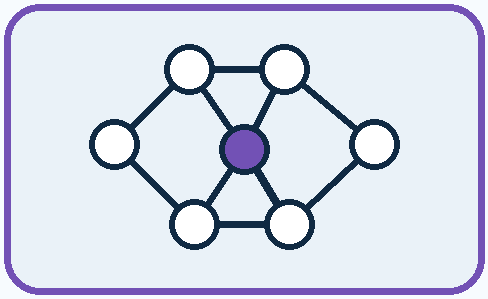
Follow a worked example

> Follow the worked trace for Find a Real Problem, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE 'Students **need** AI' is not a **problem statement**; 'students spend five minutes finding misplaced supplies after art' is observable and testable.

Step 1

STEP 1 — OBSERVE **observation**s and **interview evidence**

Step 2

STEP 2 — TRACE identify **user**, situation, **need**, and measurable effect

Step 3

STEP 3 — CONCLUDE testable **problem statement** without a preselected solution

Model answer

MODEL REASONING A strong statement names the **user**, context, observable difficulty, and desired improvement without requiring AI.

Protect privacy, avoid leading questions, and distinguish what the **user** said from your inference.

A claim to avoid

DO NOT OVERCLAIM Do not collect personal stories unrelated to the design question.

What new **evidence** could change this answer?

A closer explanation

Project choice	Boundary for this sequence	Reason
Users and setting	Fictional classroom activity requests	A clear situation to test
Information	Available minutes and quiet-room requirement	Relevant to compatibility
Action	Suggest an activity or report no match	The user retains the choice
Outside the project	Names, profiles, purchases and public posting	Not needed for this paper task

Worked explanation · Find a Real Problem

The teacher permits a paper prototype using fictional requests and an approved catalogue. We do not **need** names, home details or a history of a pupil's choices.

A request can say "10 minutes, quiet room required" without identifying anyone. Use only the request information **need**ed to test the intended **outcome**.

Worked explanation · Find a Real Problem

A clear catalogue or a short checklist may solve this problem. We will compare a proposed recommender with a simple starting method instead of assuming a more complicated system is better.

This project can use explicit rules; it does not have to fit a machine-learning model. The choice should follow the problem, available **evidence** and limits of the task.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Find a Real Problem

A fictional class loses time finding the correct craft tray.

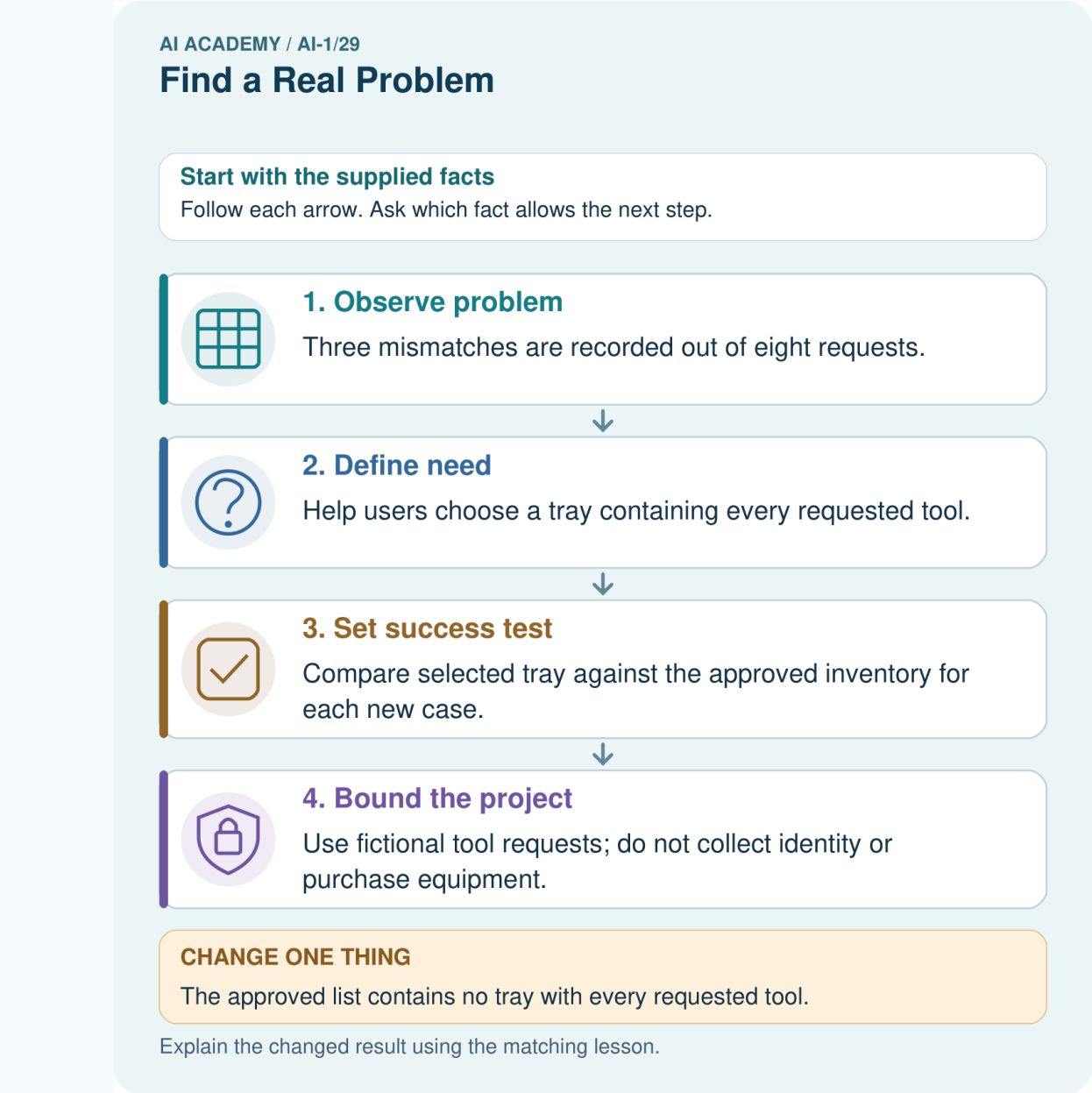
The facts we will use

Supplied facts

In eight recorded requests, three selected trays lack a required tool; five match.

Supplied facts

The teacher permits a paper finder using fictional requests and an approved tray list, with no pupil names.



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Observe problem	Three mismatches are recorded out of eight requests.
2. Define need	Help users choose a tray containing every requested tool.
3. Set success test	Compare selected tray against the approved inventory for each new case.
4. Bound the project	Use fictional tool requests; do not collect identity or purchase equipment.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The mismatch record describes a concrete **need**. A measurable goal checks whether the chosen tray satisfies the request; calling the system smart would not tell us how to evaluate it.

Start with a transparent paper baseline, and keep unsatisfied requests visible instead of selecting a near match without warning.

What this example does not establish

Keep the stated limit

Eight requests do not show the problem's frequency across the entire school or prove AI is necessary.

Keep the stated limit

Read the labelled picture: Write a one-sentence project goal and say what **evidence** would count as a failure.

Keep the stated limit

Change one thing: The approved list contains no tray with every requested tool. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Find a Real Problem claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Students need AI	technology-first claim	weak framing
Books are messy	vague judgment	needs observation
Librarian re-sorts 20 books daily	specific burden	testable need
One student opinion	useful but limited	seek more evidence

Repair two tempting claims

Challenge claim A

CLAIM A A popular technology is proof of a real problem.

Repair:

Challenge claim B

CLAIM B The first **interview** represents every **user**.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

Before building, say what would count against your idea. If a clear catalogue already produces compatible choices for all supplied cases, adding a recommender might offer little benefit on those cases.

If the recommender hides valid options or produces unexplained no-match messages, revise it or keep the simpler method. A project can succeed as an investigation even when the **evidence** favours a simpler solution.

Check the claim against the facts

Keep the ten-request **observation** log separate from later evaluation records. It motivates the problem; it is not a fair test result for a tool that has not been built.

Later weeks will introduce explicit requirements and test cases, including a reserved final set. Label plans as plans and measured results as results.

A proposed system always says “Ask the teacher”. It never suggests an incompatible activity. Has this alone demonstrated the intended benefit? Explain which additional result you would need to check

STUDENT LESSON · 8 OF 18

Practise together

Practise Find a Real Problem with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A strong project begins with a specific **user** and observable **need**.

Use a fresh example to explain **user**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: The first **interview** represents every **user**.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Suppose a team writes “All pupils hate the catalogue because it is too long.” The supplied log says nothing about every pupil’s feelings or the cause of the mismatches.

A more careful note is “Four of ten first choices were incompatible; the cause is not yet known.” A hypothesis such as “the time information may be hard to find” is something to investigate, not an established finding.

> **Write a short project brief for the activity problem. Include the users and setting, the supplied evidence, the desired outcome, information needed, one boundary and a simple alternative. Name one claim the ten-request log does not establish**

If we later investigate the cause, we could inspect where compatibility information appears in the catalogue or have a teacher organise a permitted comparison of clearer formats.

Do not quietly collect real pupils' data just because it might help. For this sequence, the supplied fictional records are enough to build and test a paper **process**.

Practise with less help: Identify claims beyond the evidence

Use workbook W2 for the connected example "A problem worth solving". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for A problem worth solving. Proposed explanation: "Using AI is itself the **outcome**". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if **need**ed.

Answer on your own before discussion. If you are unsure, say which step you **need** help with.

Explain the picture

Write a one-sentence project goal and say what **evidence** would count as a failure.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

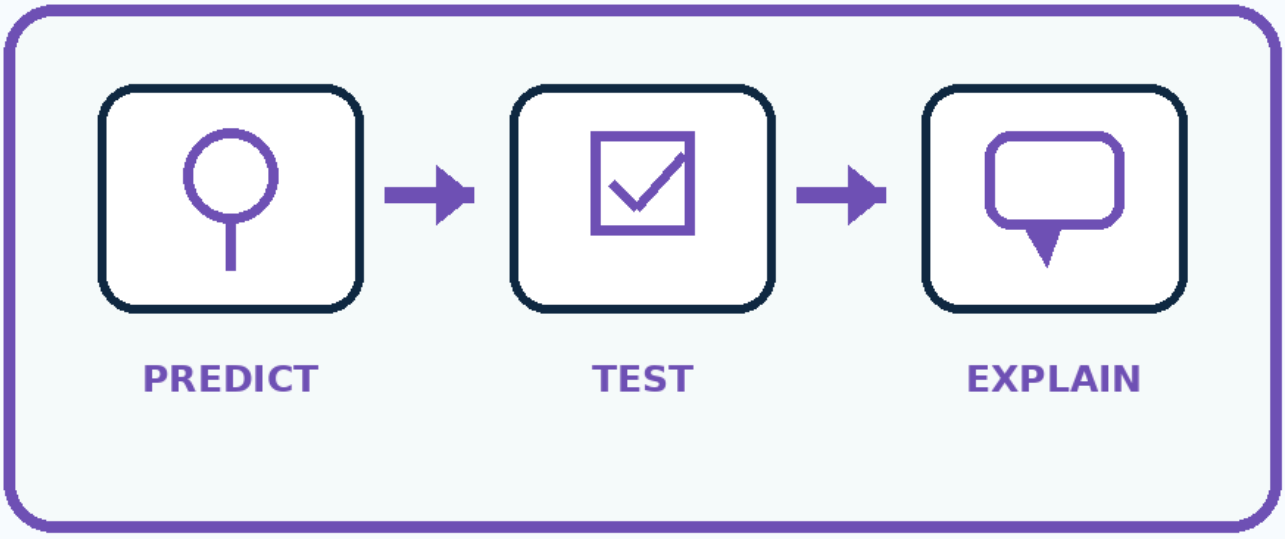
Run the investigation

> Investigate Find a Real Problem. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Need-Finding Interview

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: **Interview** a classmate and write one testable **need** statement.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

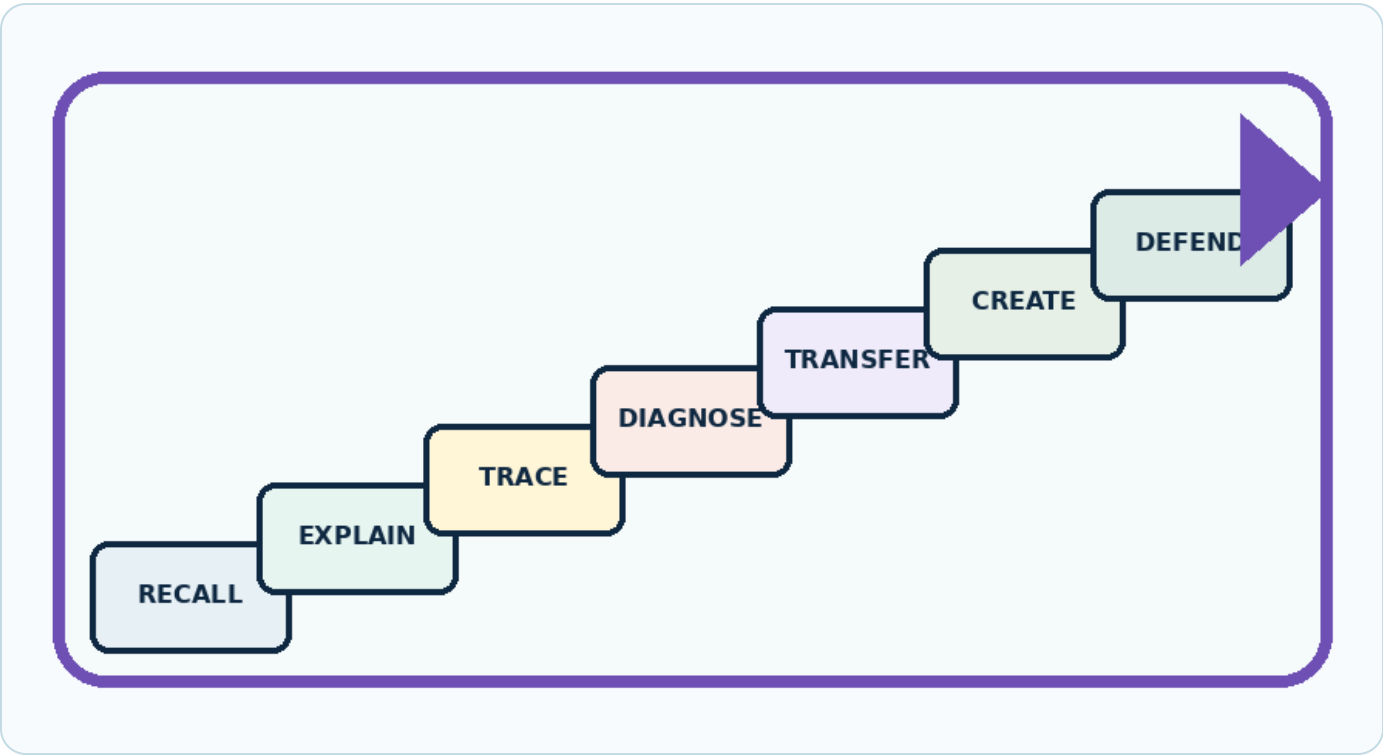
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Find a Real Problem. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **user**.

EXPLAIN A strong project begins with a specific **user** and observable **need**.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: A popular technology is proof of a real problem.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

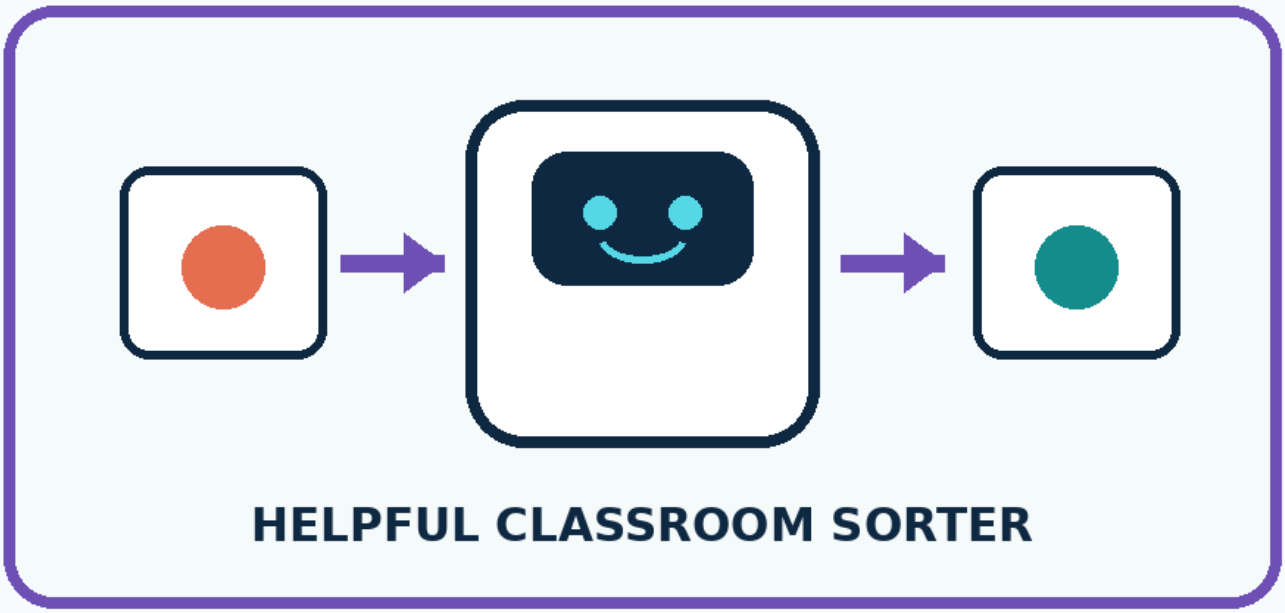
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



> THIS WEEK'S CHECKPOINT Choose the Classroom Sorter's real **user** and measurable **need**; keep an AI-free solution option open.

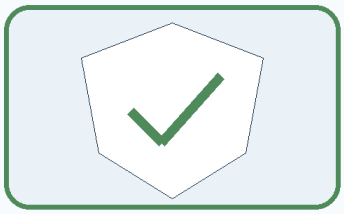
What real classroom **need** does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN Do not collect personal stories unrelated to the design question.

My project decision:

Evidence for your project

For the Helpful Classroom Sorter, retain the problem record, permitted scope, baseline and observable acceptance criteria.

STUDENT LESSON · 12 OF 18

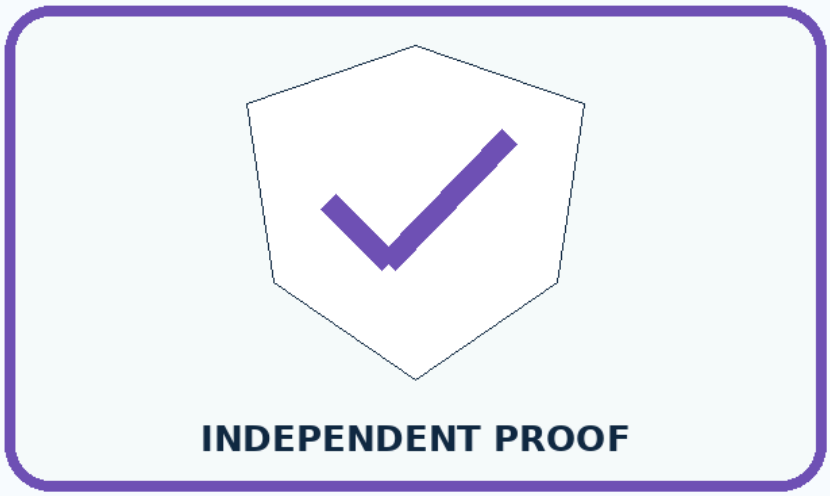
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF **Interview** a classmate and write one testable **user-need** statement

supported by an **observation** or quote.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK A strong statement names the **user**, context, observable difficulty, and desired improvement without requiring AI. Protect privacy, avoid leading questions, and distinguish what the **user** said from your inference.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Choose **Input**s and **Output**s.

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Find a Real Problem case for supported practice. Repeated examples do not become fresh assessments.

Week 29 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Shared markers are often lost because tray labels are unclear.

The question

State the **need** before selecting AI.

Worked explanation

Students **need** to return markers correctly and quickly. Test clearer labels before proposing a model.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Solve the need before selecting technology

Evidence or stage	Value or result
Problem	Markers returned to wrong trays
Possible cause	Unclear tray labels

Evidence or stage	Value or result
Simple trial	Clearer labels
Evidence	Measure correct and quick returns

Shared markers are often lost because tray labels are unclear.

State the **need** before selecting AI.

Students **need** to return markers correctly and quickly. Test clearer labels before proposing a model.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Frame a different problem independently

In a fictional equipment cupboard exercise, eight teams choose a box from a list. Three boxes do not contain the items required by the corresponding request; five do.

The teacher permits a paper box-finder using fictional item requests and an inventory. Write a brief naming the **user** difficulty, **evidence**, desired **outcome**, necessary information, an excluded action and a simple alternative.

Identify one cause or benefit that these counts do not establish.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 29 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Learners often cannot find borrowed classroom books. Define the problem before choosing an AI tool. Suggest one simple solution to compare with AI.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Choose the Classroom Sorter's real **user** and measurable **need**; keep an AI-free solution option open.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Students spend three minutes finding returned books. Compare shelf labels with an AI locator. Define one fair success test before choosing a solution.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Find a Real Problem, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional class loses time finding the correct craft tray.

By the end, I can	How I will show it
Describe a specific user difficulty using supplied evidence without inventing its cause	Explain the deciding step, show a trace or test, and state one limit.
Define a useful outcome and a way to observe it before choosing a solution	Explain the deciding step, show a trace or test, and state one limit.
Choose a small project with clear boundaries, permitted information and a simple alternative	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Find a Real Problem without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Describe a specific **user** difficulty using supplied **evidence** without inventing its cause

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional class loses time finding the correct craft tray.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Use fictional tool requests; do not collect identity or purchase equipment.

1. Bound the project

2. Observe problem

3. Set success test

4. Define **need**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if **need**ed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Bound the project Use fictional tool requests; do not collect identity or purchase equipment. The mismatch record describes a concrete **need**.

A measurable goal checks whether the chosen tray satisfies the request; calling the system smart would not tell us how to evaluate it.

Start with a transparent paper baseline, and keep unsatisfied requests visible instead of selecting a near match without warning.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The approved list contains no tray with every requested tool. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. The finder should report no suitable tray rather than guess.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if **need**ed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The finder should report no suitable tray rather than guess. Success includes honestly identifying unsatisfied requirements.

Repair a missing building block

Review Humans Set Boundaries (ai-1/28). Goal: Distinguish a system's capability, the correctness of its **output** and permission to take an action. Short practice: Which original proposal is authorised?

Why do the other two **need** separate approval?. Return to this week and explain what you changed.

Review Prompts Are Design Instructions (ai-1/26). Goal: Write a prompt with a clear task, supplied **evidence** and checkable **output** requirements.

Short practice: What must happen if B instead conflicts with another equally authoritative duration record?. Return to this week and explain what you changed.

Review Fair for Whom? (ai-1/23). Goal: Compute a correct-out-of-assessed fraction and explain its denominator. Short practice: Which group loses performance under N, and can total improvement erase that fact?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

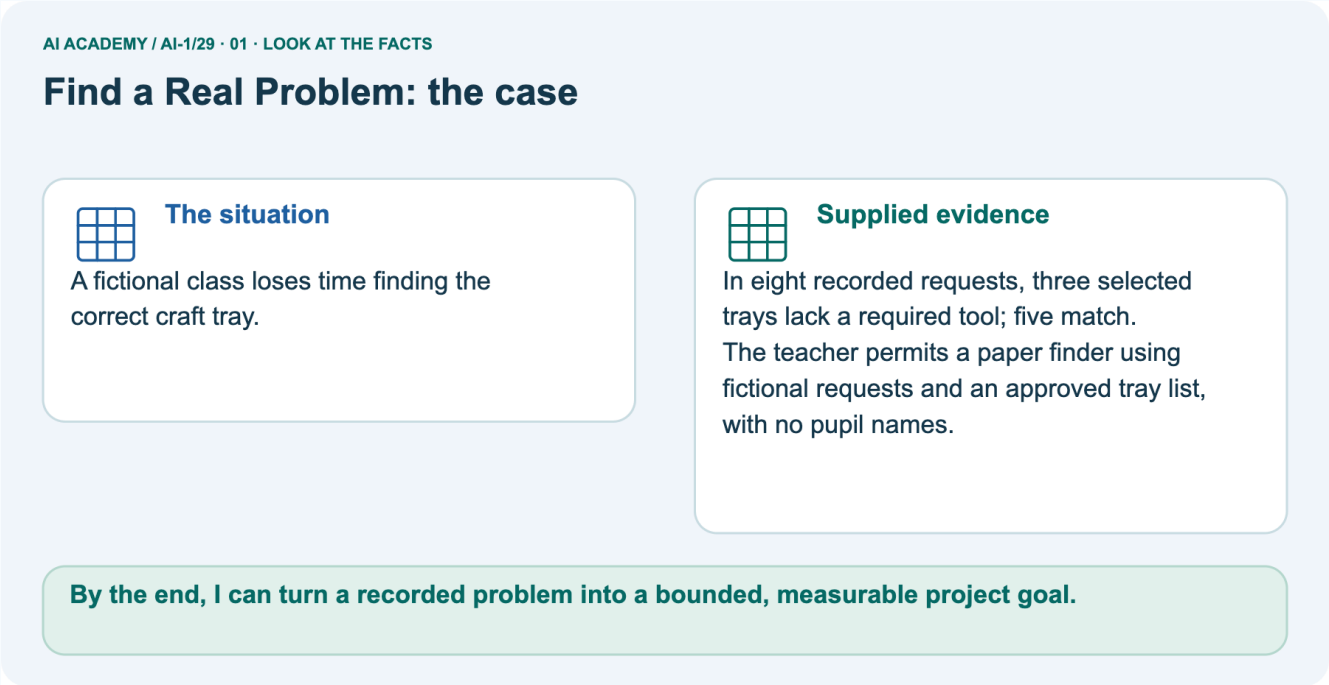
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Find a Real Problem graphic using its labels, deciding evidence and relevant limitation.

01. Find a Real Problem: the case



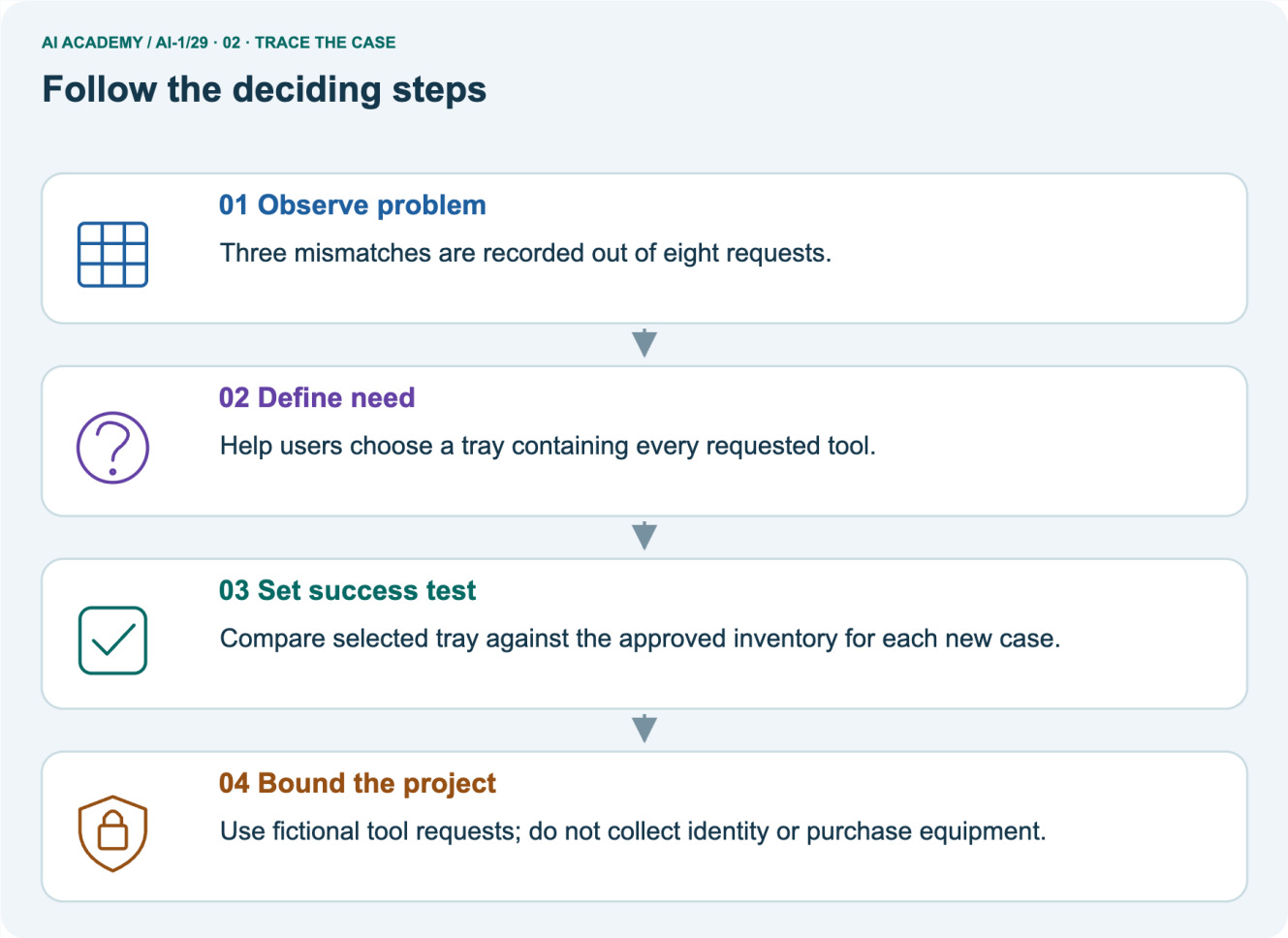
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional class loses time finding the correct craft tray. Supplied facts: In eight recorded requests, three selected trays lack a required tool; five match.

The teacher permits a paper finder using fictional requests and an approved tray list, with no pupil names.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Observe problem: Three mismatches are recorded out of eight requests. → Define **need**: Help **user**s choose a tray containing every requested tool. → Set success test: Compare selected tray against the approved inventory for each new case. → Bound the project: Use fictional tool requests; do not collect identity or purchase equipment.

> Point to each numbered stage and explain which recorded **input** it uses.

03. From an observed difficulty to a testable goal



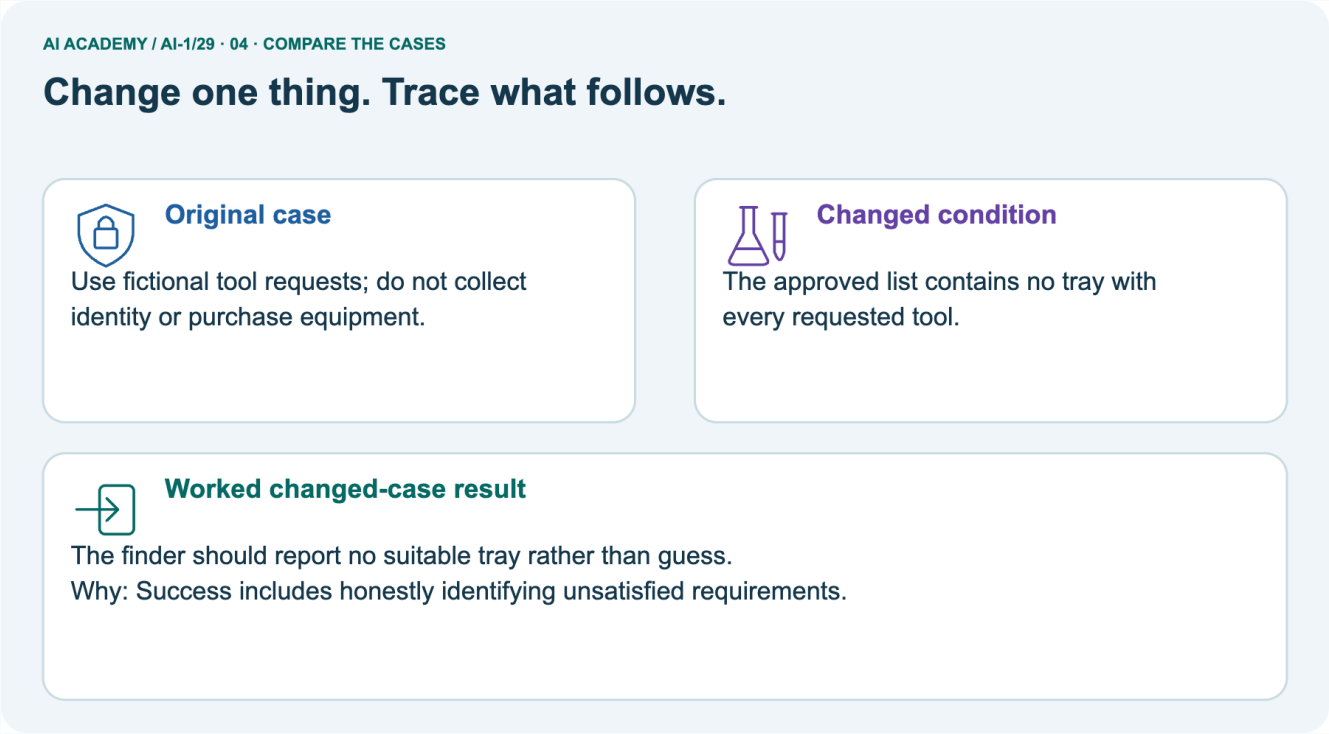
Follow the labelled picture with the matching explanation. Open the vector diagram

From an observed difficulty to a testable goal. 4 of 10 choices mismatch | Some re-selection occurred | Cause is not established. **Need** a compatible choice | Check time and conditions | **User** can still decline.

Paper suggestion **process** | Small permitted prototype | Benefit **need**s a comparison. Measure fit and coverage | Observe useful **outcome**s | Do not hide no-result cases

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



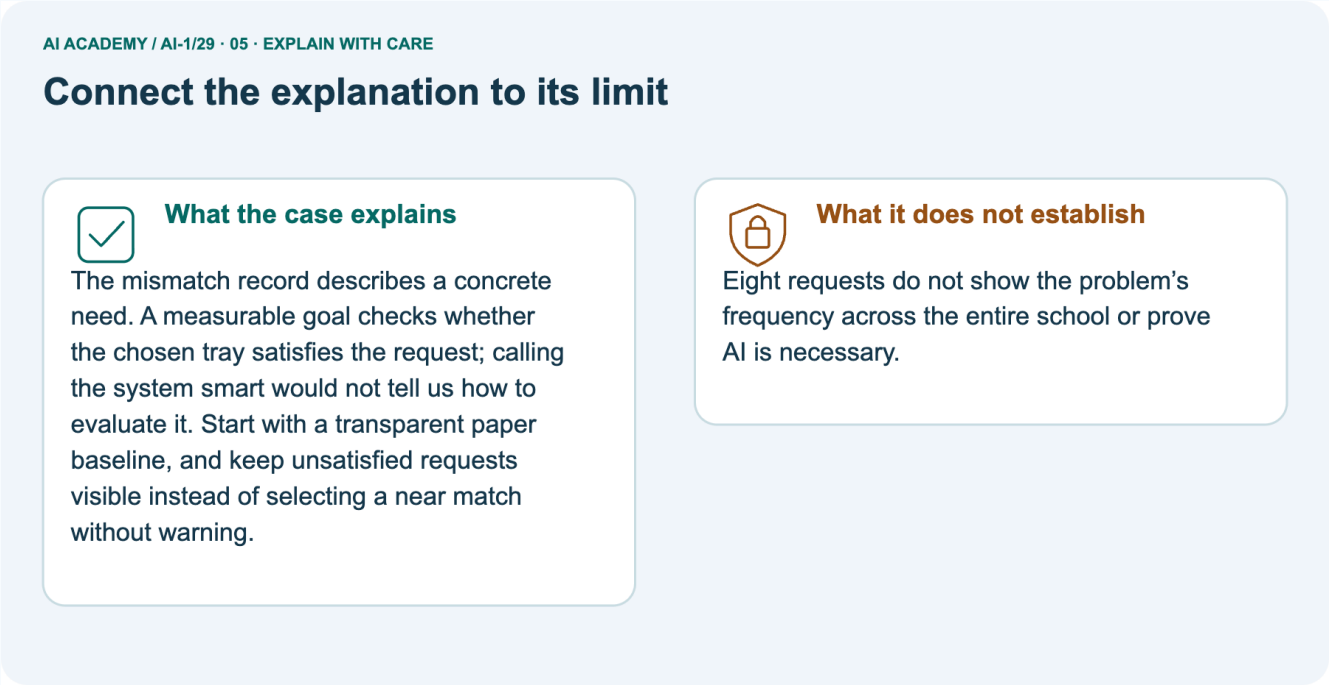
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Use fictional tool requests; do not collect identity or purchase equipment. Changed condition: The approved list contains no tray with every requested tool.

Expected result: The finder should report no suitable tray rather than guess. Reason: Success includes honestly identifying unsatisfied requirements.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

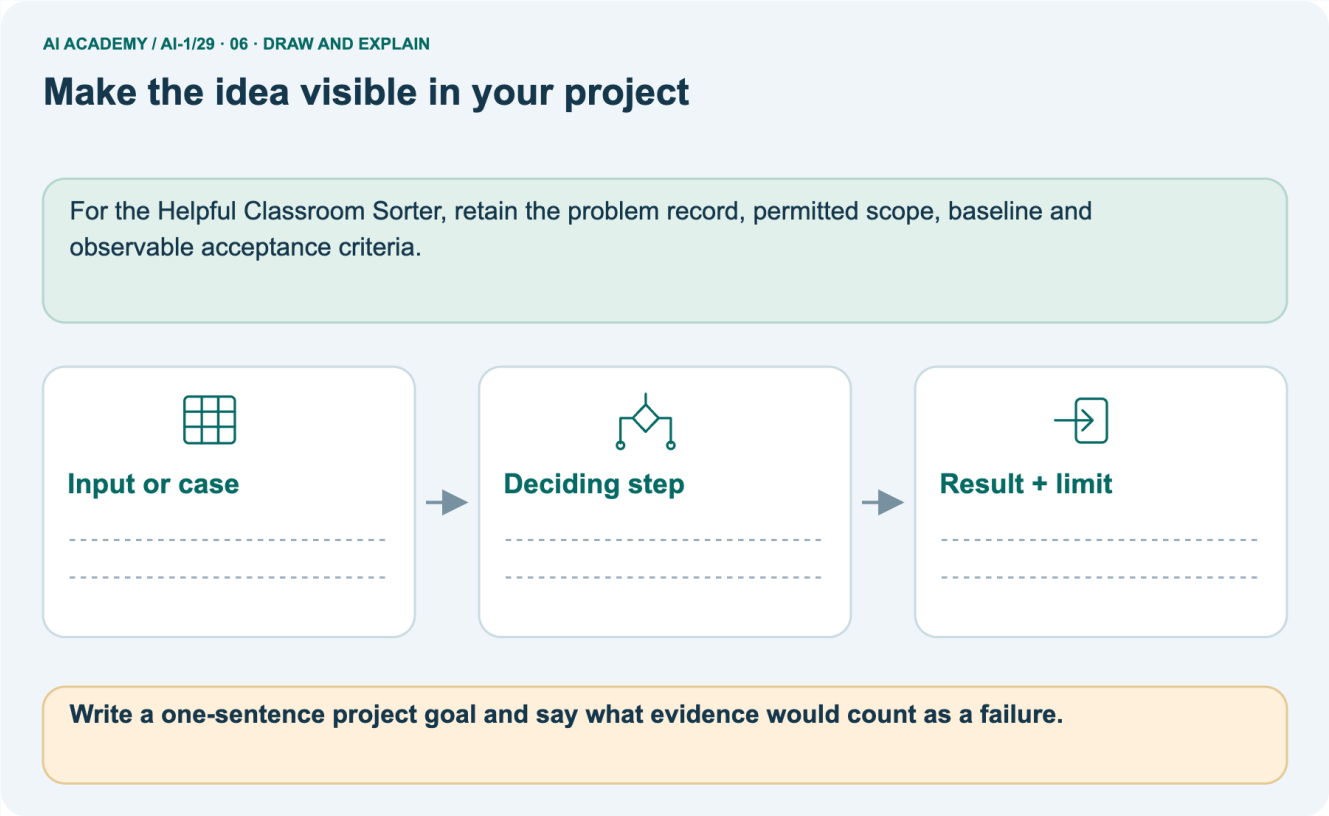
The mismatch record describes a concrete **need**. A measurable goal checks whether the chosen tray satisfies the request; calling the system smart would not tell us how to evaluate it.

Start with a transparent paper baseline, and keep unsatisfied requests visible instead of selecting a near match without warning.

Limit: Eight requests do not show the problem's frequency across the entire school or prove AI is necessary.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Helpful Classroom Sorter, retain the problem record, permitted scope, baseline and observable acceptance criteria. Draw your **input**, deciding step, result and limit.

Supported question: Write a one-sentence project goal and say what **evidence** would count as a failure.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.